

REVIEW

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# Reconstructive spectrometers: hardware miniaturization and computational reconstruction

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## Abstract

Spectrometers serve as indispensable analytical tools across chemistry, materials science, environmental monitoring, medical diagnostics, and beyond. The emergence of reconstructive spectrometers represents a transformative leap in spectral analysis, combining miniaturized encoding hardware with advanced computational algorithms to revolutionize conventional approaches. These devices encode unknown spectral data into measurable signals, for which sophisticated algorithms then decode to reconstruct the original spectrum with high fidelity—all achieved within an ultra-compact footprint. In this review, we first establish the mathematical foundations governing spectral encoding and decoding. We then provide a detailed analysis of encoding strategy and state-of-the-art decoding techniques, followed by recent breakthroughs in hardware design for optimized spectral reconstruction systems. Finally, we address key challenges and future opportunities, offering insights into how reconstructive spectrometers may redefine spectroscopy beyond traditional laboratory settings.

**Keywords** Decoding algorithms, Inverse design, Joint optimization, Miniaturization, Reconstructive spectrometers

## 1 Introduction

Spectrometers are sophisticated analytical instruments designed to measure the interaction of light with matter across diverse ranges of wavelengths. Traditional spectrometers disperse light into its component wavelengths to provide detailed information such as chemical composition, molecular structure, and surface texture. These devices play a pivotal role in numerous scientific disciplines and industries, serving as indispensable tools for chemical analysis, materials characterization, environmental monitoring, and biomedical diagnostics [1, 2]. However, they typically have large form factors, complex and cumbersome designs, and are expensive, limiting their accessibility and application scopes. Recent advancements in miniaturization and computational capabilities have enabled the development of compact, portable, and cost-effective spectrometers, which offer the advantage of portability while achieving competitive performance such as high resolution and accuracy.

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These miniaturized versions not only expand the reach of spectroscopic techniques to field-based, real-time applications, but also open opportunities for integration into consumer electronics and handheld devices, thereby unlocking significant market potential and enabling broader adoption across diverse sectors [3–8].

In the realm of miniaturized spectrometers, many innovative designs have emerged, each leveraging unique principles to achieve compactness while maintaining competitive performance in targeted application scenarios. These designs can be broadly categorized into four main types: miniaturized dispersive optics, narrow-band filter systems, Fourier transform interferometers, and reconstructive spectrometers [9]. Among these, the reconstructive approach stands out as particularly promising. It typically comprises a hardware encoder (e.g., an array of detectors with distinct spectral responses) and an algorithmic decoder, where the former compresses and encodes the spectral information into digital signals, while the latter recovers the unknown spectrum from the recorded signals [10]. By harnessing the power of modern computational techniques, including compressive sensing and machine learning, reconstructive spectrometers can achieve good performance, such as high spectral resolution and accuracy, while maintaining a small physical footprint [11–15].

On the other hand, recent advances in nanophotonics and materials science have enabled the creation of highly efficient and compact detectors. These detectors can be engineered to have specific spectral responses, allowing for precise and reliable spectral decoding. For example, a composition-gradient alloyed semiconductor nanowire can be divided into multiple sections, each with a slightly different alloy composition, resulting in varying spectral responses along its length [16]. Another notable breakthrough is the application of tunable photodetectors in spectrometers. Through thermal or electrical tuning, the spectral response can be modulated on a single detector, utilizing materials such as black phosphorus [17], van der Waals (vdW) junctions [18–21] and p–n junctions [22, 23], thereby advancing miniaturization toward the realization of a spectrometer-on-a-pixel. Building on these innovations, the performance of reconstructive spectrometers is influenced by two key factors: the design of the hardware encoder, and the effective implementation of the algorithmic decoder. We next examine these critical components in sequence.

First, the hardware encoder, which can be a tunable spectral component [17–23] or an array of modulating elements [24–28] with distinctive spectral responses, is crucial for capturing the necessary information for spectral reconstruction. Minimizing the correlation [24] between the spectral responses of the detectors is

advantageous, as it maximizes the diversity of detection outcomes. Furthermore, the design and fabrication of these detectors must be optimized to balance performance metrics including resolution, spectral range, and sensitivity within miniaturization constraints [25–30]. However, manufacturing processes are inherently subject to fabrication tolerances and process variations, which can affect the performance of the detectors [31, 32]. To address this, researchers simulate potential deviations introduced during the actual manufacturing phase, narrowing the gap between design and real devices. Additionally, the employment of techniques such as adjoint sensitivity analysis [33] allows researchers to assess the sensitivity of target performance to design parameters and proactively eliminate structures that are highly sensitive to these variations. This approach ensures that the designed encoder exhibits a certain degree of robustness to fabrication uncertainties, thereby enabling consistent and reliable spectral reconstruction even under less-than-ideal fabrication conditions.

Second, the algorithmic decoder is responsible for processing the raw signals collected by the hardware encoder and approximating the original spectrum. This involves solving a complex inverse problem, typically formulated as a linear equation system. Machine learning techniques, particularly deep learning algorithms, have shown great promise in enhancing the accuracy and robustness of spectral reconstruction [34–38]. These algorithms can learn intricate patterns from large datasets and possess exceptional computational efficiency, thereby standing out as excellent candidates for decoders. To enhance the generalization capability of the model, embedding prior physical information (e.g., through physics-informed learning [39]) within the decoding algorithm has been considered as a strategy to reduce the over-reliance on training data and also minimizes unnecessary computations towards unphysical solutions, thereby improving the efficiency, accuracy, and robustness of the model. To facilitate the deployment of decoding algorithms on mobile platforms, the development of lightweight algorithmic models has also drawn significant attention [40].

By optimizing both the hardware encoder and the algorithmic decoder, researchers can significantly enhance the performance of reconstructive spectrometers. However, the true potential of these systems lies in the joint optimization of these two components. The hardware encoder and the algorithmic decoder are not independent entities; rather, they function synergistically to achieve high-resolution and accurate spectral reconstruction. The interplay of these components dictates overall system performance, making coordinated hardware-software co-design essential for the development of next-generation spectrometers. This synergy

is particularly strengthened by recent advances in deep learning, which enable simultaneous learning of physical parameters of the encoder schemes and decoding algorithms. After training on datasets, such models can generate a fully matched encoder and decoder pair, exhibiting excellent spectral reconstruction performance and possessing superior tolerance to uncertainties such as manufacturing errors and measurement noise.

Distinct from prior comprehensive review [9] that centered on all miniaturized spectrometers, and building upon previous studies that predominantly focused on the decoder [41, 42], this review provides a unique perspective by emphasizing the joint hardware-algorithm optimization in reconstructive spectrometers, as well as the emerging role of low-dimensional materials and structures. We start by outlining the fundamentals, including mathematical models, and various encoding strategies. Next, we examine decoding algorithms, covering iterative reconstruction and deep learning-based rapid inversion. We then explore encoder design, including forward modeling, inverse design, and end-to-end system design. Finally, we conclude with a summary and future perspectives.

## 2 Fundamentals of reconstructive spectrometers

### 2.1 Mathematical models for encoding and decoding

The signal generated at the detector upon light incidence can be described mathematically as:

$$I = \int_{\lambda_1}^{\lambda_2} R(\lambda) \cdot S(\lambda) d\lambda, \quad (1)$$

where  $R(\lambda)$  represents the spectral response under a specific wavelength  $\lambda$ ;  $I$  the corresponding measured signal, and  $S(\lambda)$  the input light intensity at  $\lambda$ . The upper and lower limits of integration define the wavelength range of the detector. This equation essentially describes an encoding process in which spectral information is converted into light intensity. It should be noted that  $R(\lambda)$  may depend on the input light intensity due to, for example, nonlinear effects of optical elements preceding the detector or the saturation effects of the detector itself [22]. However, such cases are rarely reported and are not the focus of this review. Equation (1) can be discretized and expressed in a matrix form as:

$$\mathbf{I} = \mathbf{R} \cdot \mathbf{S}. \quad (2)$$

Here,  $\mathbf{R}$  represents the response matrix with dimensions  $m \times n$ .  $\mathbf{I}$  and  $\mathbf{S}$  are the vectors with dimensions  $m$  and  $n$ , respectively. Unlike conventional methods that directly measure the intensity of each wavelength, reconstructive spectroscopic devices aim to recover the spectral information with fewer measurements, leading to an

underdetermined equation system for compressed sampling ( $m < n$ ). To form the response matrix  $\mathbf{R}$ , two general ways are space domain sampling and time domain sampling, with rows of  $\mathbf{R}$  corresponding to the spectral modulation from multiple channels and multiple states with one channel, respectively. The key to the successful reconstruction from limited sampling with a compression ratio of  $n/m$ , which appears to violate the Nyquist sampling theorem, lies in the sparsity of the spectral signal. This implies that the target spectrum comprises only a few significant wavelength components as described by:

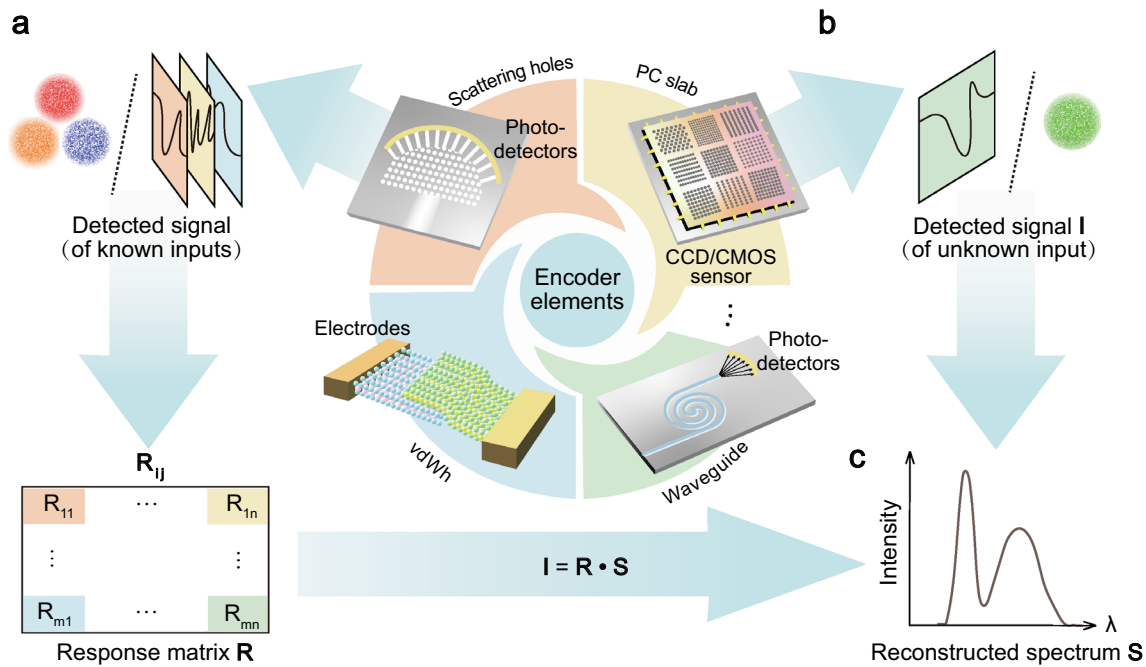
$$\mathbf{S} = \varphi \alpha, \quad (3)$$

where  $\mathbf{S}$  represents the spectrum,  $\varphi$  the orthogonal basis in which the spectrum is sparse (note, it can also be an identity matrix), and  $\alpha$  the weight coefficient vector. The transformed domain mentioned here can be intuitively interpreted as a set of designed sparse bases, i.e., the columns of the transformed matrix. In other words, a linear combination of a small subset of these bases can represent the unknown spectrum. Therefore, we can recover spectral information with fewer measurements based on compressive sensing (CS) theory, thereby establishing a framework for designing efficient decoders that exploit sparsity in the transformed domain.

Reconstructive spectrometers operate through three sequential stages: calibration, measurement, and reconstruction (Fig. 1). The process begins with calibrating the spectral response of the encoder to establish the critical response matrix  $\mathbf{R}$ , achieved by exposing the device to light with known spectral profiles (Fig. 1a). During measurement (Fig. 1b), unknown spectrum of light is irradiated onto the detectors, and the corresponding output signal  $\mathbf{I}$  is captured. Finally, in the reconstruction stage (Fig. 1c), the input spectrum is recovered by solving the linear system  $\mathbf{I} = \mathbf{R} \cdot \mathbf{S}$  (Eq. 2), where  $\mathbf{S}$  represents the reconstructed spectrum. This inversion process leverages the pre-calibrated matrix  $\mathbf{R}$  and measured signal  $\mathbf{I}$ , effectively translating compressive measurements into high-fidelity spectral data through sparse encoding principles.

### 2.2 Encoding strategy

Compressed sensing theory dictates that an effective spectral response must exhibit two essential characteristics: (1) sufficient randomness and spectral variability, manifested through narrow autocorrelation bandwidth (as quantified by the FWHM of the signal's self-similarity profile), and (2) minimal cross-correlation (i.e., low inter-channel similarity) between spectral channels [43, 44]. This allows each encoder to capture adequate information in the wavelength domain and ensures a high degree of independence between different channels, thereby achieving sufficient sampling while improving



**Fig. 1** Schematic of the working principle of computational spectrometers. **a** Learning process (also known as the calibration process). Various optical elements act as encoders. The response matrix  $R$  of the encoders is measured via diverse known spectra. **b** Measurement process. The output signal  $I$  is detected under an unknown incident light. **c** Reconstructed spectral information  $S$  by computational methods with the data obtained in the above two processes

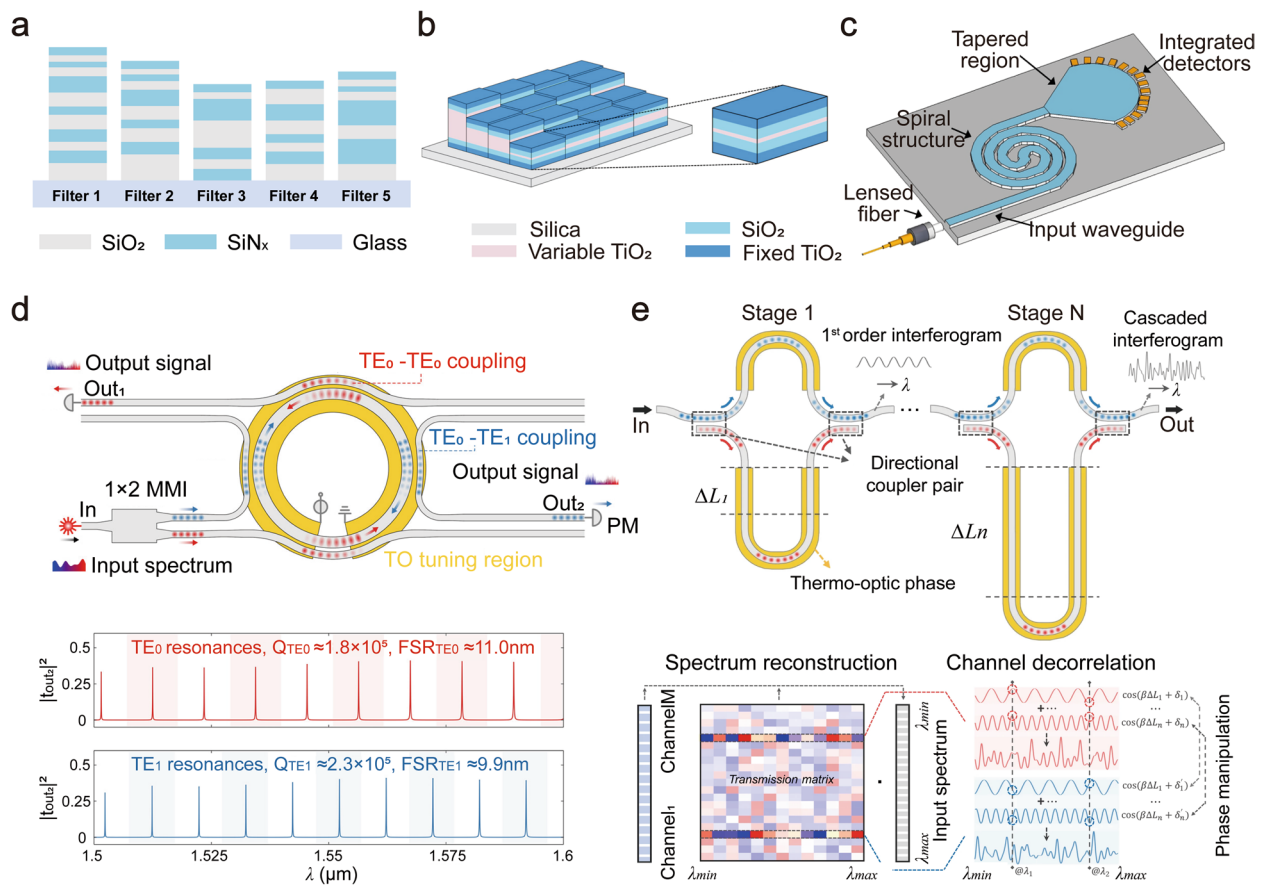
reconstruction accuracy. In addition, an ideal encoder is expected to exhibit sufficient measurement bandwidth, robust resistance to measurement noise (reflected by a low condition number of the response matrix), and high efficiency in utilizing optical energy. These characteristics ensure superior performance of spectrometers in terms of sensitivity, stability, and dynamic range. It is important to note that the field of spectrometers currently lacks a rigorous framework for evaluating encoder performance. On one hand, performance metrics related to the response matrix, such as correlation coefficients between matrix rows, are insufficient to fully assess the capability of the encoder to reconstruct unknown spectral features [45]. On the other hand, certain parameters, such as dynamic range, are challenging to characterize using the response matrix alone. Therefore, establishing a standardized quantitative framework to describe the performance of spectrometers represents a critical issue that needs to be addressed in the future.

In recent years, researchers have developed various photonic structures with sophisticated spectral modulation to explore encoding schemes for reconstructive spectrometers. Based on different encoding strategies, we categorize the mainstream miniature reconstructive spectrometers into four groups: (i) interference, (ii) scattering, (iii) diffraction, and (iv) band-gap modulation.

### 2.2.1 Interference

Various optical interference elements have been demonstrated as effective encoders for reconstructive spectrometers, including multilayer thin-film (MTF) filters [25, 46, 47], Fabry-Pérot (FP) filters [27, 48, 49], waveguides [31, 50], microring resonators (MRRs) [15, 51–55], and Mach-Zehnder interferometers (MZIs) [52, 56], etc. These components achieve complex spectral modulation by exploiting constructive and destructive interference between reflected and transmitted light at engineered material interfaces. The transmittance of MTFs, specifically, is fundamentally determined by the wavelength and angle of incident light, as well as the designed thickness and material composition of each layer (e.g., air/alternating layers of silicon nitride and silicon dioxide/glass substrate, as shown in Fig. 2a). The spectral diversity is engineered through lithographic patterning. While maintaining uniform material composition across layers, photoresist masks enable selective layer deposition or etching to create spatially varying configurations [25, 46]. This structural modulation alters optical path differences, thereby generating tailored spectral responses. Recent advancements demonstrate that integrating MTFs with metasurfaces can synergistically enhance spectral control through hybrid interference-diffraction effects [57]. In contrast, FP systems derive their spectral tuning





**Fig. 2** Interference optical elements of miniaturized reconstructive spectrometers. **a** Membrane system of MTF structure with fixed thickness, and **b** membrane system of F-P structure with gradient thickness. **c** A compact spectrometer based on a spiral waveguide. **d** Schematic of a single micro-ring resonator spectrometer and the calculated transmission responses of the dual modes. **e** Schematic of a programmable spectrometer using multiple unbalanced MZI stages and its working principle. **b** Reprinted from ref. [49] with permission from ACS. **c** Reprinted from ref. [31] with permission under CC BY 4.0. **d** Adapted from ref. [53] with permission under CC BY 4.0. **e** Adapted from ref. [56] with permission under CC BY 4.0

capability primarily from precise thickness control of individual dielectric or polymer layers (Fig. 2b).

Photonic structures with higher quality factors (Q), such as waveguides [31, 50], MRRs [15, 51–55], and MZIs [52, 56], are mainly fabricated on silicon-on-insulator (SOI) or silicon nitride (SiN) platforms, with SOI providing a superior index contrast and a smaller footprint while SiN provides better stability against temperature fluctuations. Emerging functional substrates like thin-film lithium niobate are also gaining attention [58, 59]. These structures typically exploit the sensitivity of the optical path difference (OPD) to wavelength, thereby causing light of different wavelengths to interfere and produce distinct detection signals. In multimode waveguides, the characteristic interference pattern emerges from modal interactions, enabling the reconstruction of the spectrum. Therefore, the waveguide is usually designed with a stratified structure [44] or helical

geometry [31] to ensure adequate optical path length difference over a certain wavelength range within a compact device (Fig. 2c). However, the resulting elongated optical path and curved structure of the multimode waveguides also introduce a certain level of loss. Alternatively, a single MRR supporting dual-mode resonances offers a more compact solution [53], with resonant wavelengths adjustable through thermal regulation (Fig. 2d). To cover a broader spectrum with high efficiency, an MRR with a larger free spectral range [54] or a cascade arrangement of MRRs [51, 52] can be employed. Similarly, MZIs can also modify the optical path length through thermal or electrical tuning. By cascading MZI optical switches, it is possible to freely switch the optical path length, thereby achieving an exponential increase in sampling channels within a compact footprint [51, 56] (Fig. 2e). Another class of high-Q spectral response, characterized by significant disorder and high sensitivity to initial

conditions, has emerged as a highly promising approach arising from the “chaotic behavior” generated by multi-mode mixing and cavity resonance effects. This response can be realized through dense waveguide arrays [60], where thermally tuned temporal speckle overcomes the limitations of traditional spatial modes and breaks the capacity constraints of multimode interference; through heteronuclear photonic molecule systems [61], in which mode hybridization and spatial symmetry breaking enable whispering-gallery modes interfering, generating disordered supermodes that suppress spectral periodicity and synergistically achieve ultra-high resolution and large bandwidth; and additionally, chaotic cavity structures employing boundaries shaped as Pascal’s limaçons [62], etc. Collectively, these approaches advance computational spectrometers towards becoming “smaller and more comprehensive”.

The aforementioned structures also present certain challenges. While reconstructive spectrometers based on filter arrays and Fabry-Perot cavities have matured in their processes and are easily integrated with CMOS [63], the sensitivity to incident light angles can significantly impact spectrometer performance. The angular sensitivity undoubtedly elevates the precision requirements of optical paths during calibration processes, inevitably introducing extraneous errors in practical applications. The filter configurations inherently induce a certain loss in light flux for detection systems, consequently affecting system dynamic range and sensitivity. Furthermore, the alignment of the filter with the photodetector imposes additional manufacturing complexities, introducing spatial crosstalk to spectrometer systems. However, for spectral imaging, especially in systems operating at video-level imaging rates, filter-based reconstructive spectrometers still retain advantages in cost and system configuration.

On-chip spectrometers based on structures such as waveguides, MZIs, and MRRs are poised to become a significant branch of reconstructive spectrometers. These spectrometers can integrate various components within a single chip and are compatible with the mature CMOS technologies [64]. Furthermore, through on-chip switching networks, the spectrometers can achieve active optical path configuration rather than spectral splitting. However, these increasingly sophisticated on-chip interconnections lead to higher power consumption. In certain compact on-chip configurations, the loss of optical power within components, as well as the crosstalk between different components, introduces additional complexity to the calibration of the devices (e.g., necessitating additional measurements of coupling coefficients between components). Moreover, for electro-optic modulation, the availability of suitable materials is relatively

limited, which adds to the cost of device fabrication, and also imposes certain constraints on the operational spectral range.

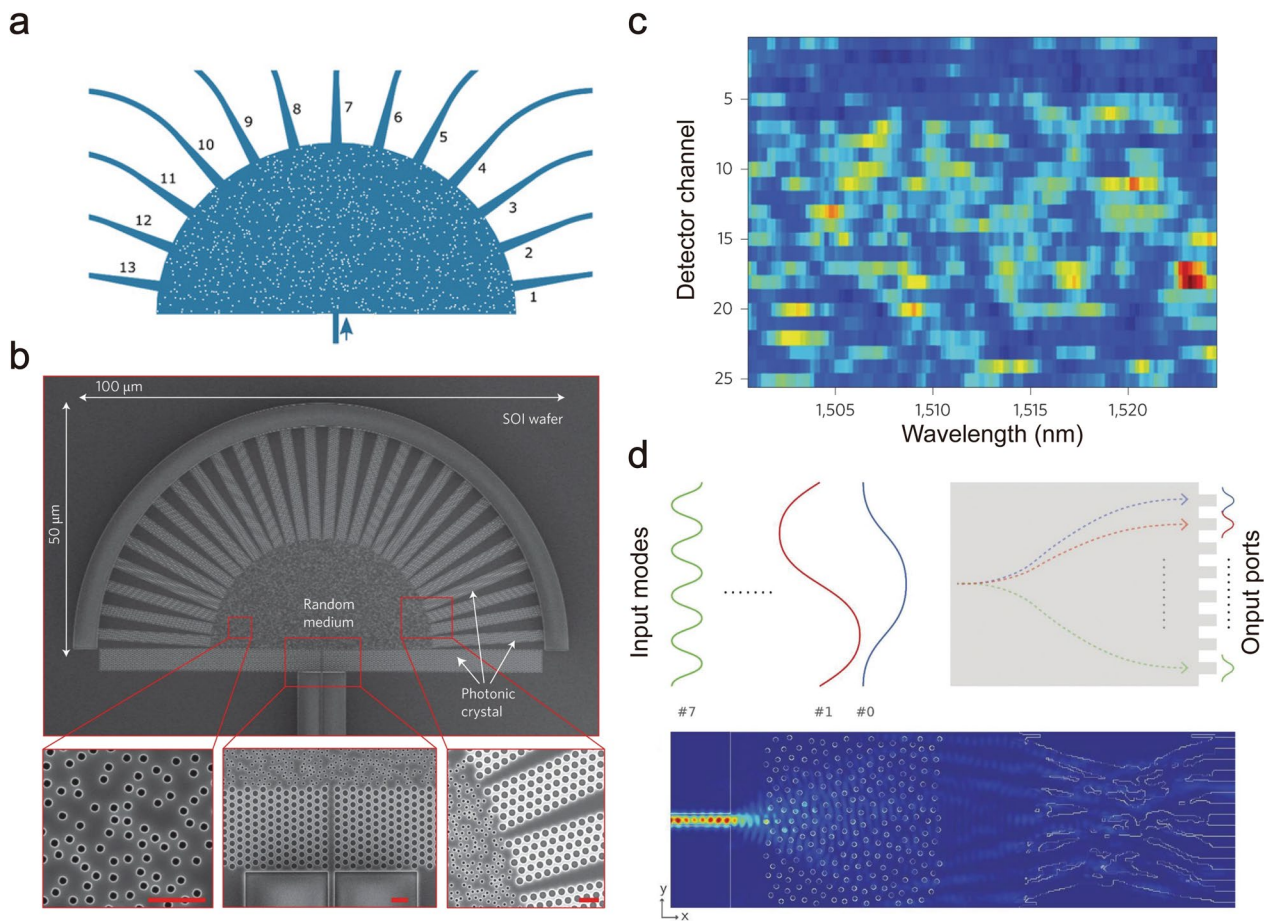
### 2.2.2 Scattering

In contrast to meticulously designed structures, disorders can also be harnessed in developing miniaturized spectrometers. The fundamental mechanism of these spectrometers relies on light traversing a scattering medium, which facilitates effective mode mixing by shortening the mean free path of light transport. Therefore, the scattering medium creates complex and quantifiable light patterns (modes). In this context, disorders are intentionally introduced in the encoder to amplify scattering effects (Fig. 3a). As shown in Fig. 3b, broadband light enters the tailored structure, for instance, air holes, where strong scattering fosters an exceptionally efficient blending of photon modes within a confined space [32, 65–67]. As the light propagates through the disordered scattering medium, it produces a distinct pattern with a unique wavelength-dependent intensity distribution at the output, which can subsequently be utilized for reconstruction purposes (Fig. 3c). Nevertheless, spectrometers based on disordered scattering typically require a substantial scattering area. This limitation can be mitigated by integrating an inverse-designed mode decomposer, which can direct specific modes of the incident light into their designated ports to achieve efficient mode redirection [66] (Fig. 3d).

While intense scattering promotes highly efficient mode mixing within a compact area, minimizing light scattering leakage to combat decreased transmission remains challenging. On one hand, the enhancement of resolution necessitates an increase in the OPD of the system; on the other hand, larger OPD can lead to increased scattering losses in the out-of-plane direction, thereby limiting the sensitivity. Previous literature has demonstrated that compared to fully random disordered structures, the introduction of partially random structures can mitigate out-of-plane scattering, allowing more light to reach the detectors [66]. Furthermore, the stochastic nature of scattering structures raises the bar for manufacturing stability and reproducibility, posing a hurdle for large-scale production. Random structures may exhibit heightened sensitivity to interference and noise [32], thus maintaining environmental stability during the calibration phase is also a pertinent issue that needs to be addressed.

### 2.2.3 Diffraction

As is for diffractive elements, periodic nano- and microstructures exhibit exceptional dimensional compactness, which is crucial for miniaturizing spectral devices. The



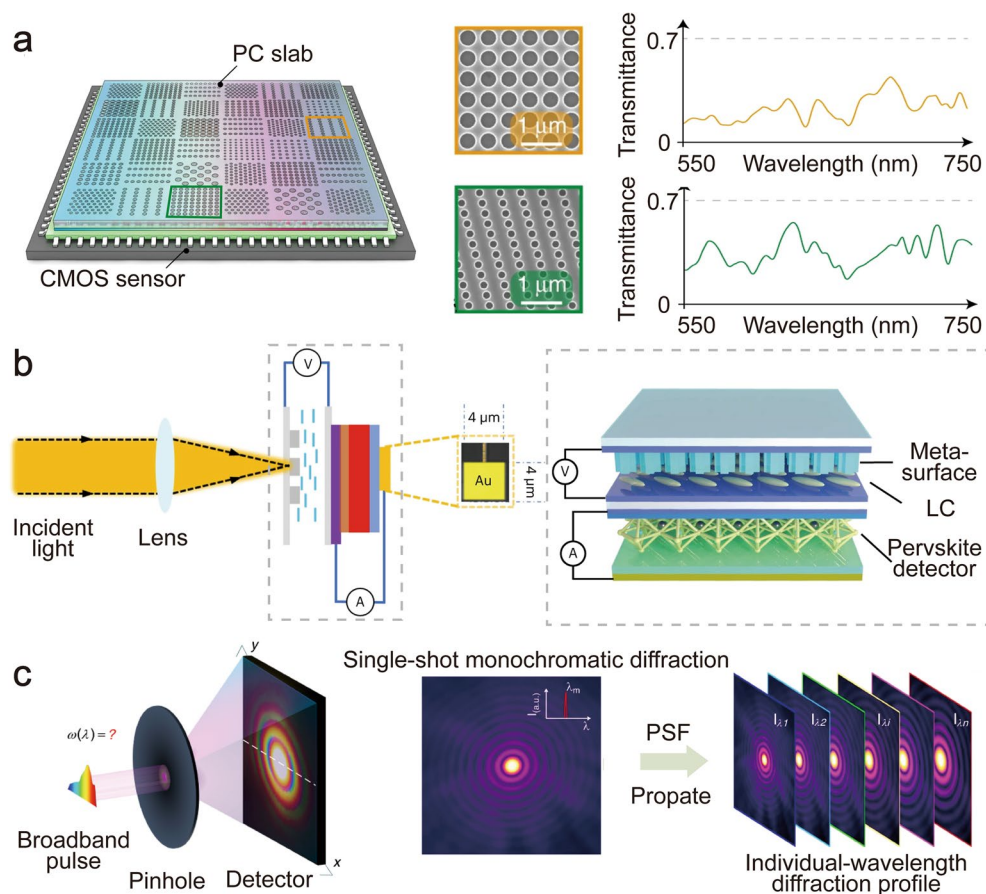
**Fig. 3** Scattering optical elements of miniaturized reconstructive spectrometers. **a** A random scattering spectrometer. **b** Scanning electron microscope (SEM) images of a random scattering structure. **c** The transmission matrix of the random scattering spectrometer. **d** Definition of the inverse-designed mode decomposer and the simulated electric field intensity profile of the spectrometer. **a** Reprinted from ref. [65] with permission under CC BY 4.0. **b, c** Reprinted from ref. [32] with permission from Springer Nature. **d** Reprinted from ref. [66] with permission from John Wiley and Sons

Bragg diffraction in photonic crystals (PC) and the engineered diffraction in metasurfaces both require precisely patterned nanostructures, which enable comprehensive control of light's amplitude, phase, polarization, and orbital angular momentum [68]. In these devices, PCs facilitate the coupling of free-incidence light with lateral propagation modes, leveraging periodic nanostructures to enhance light path reflection [69, 70] (Fig. 4a). Adjustments to lattice constants and apertures alter the trajectory and reflection of light, leading to complex optical phenomena such as transverse light bouncing [71], longitudinal Fabry-Pérot resonances [72], guided resonances [73], and Fano interference [74], collectively contributing to the richness of spectral characteristics.

In the context of metasurfaces, light modulation is achieved by varying the material, geometry, size, and period of the periodic patterns. Existing research has showcased a diverse array of metasurface patterns,

including cross-bars [11, 26], square [26], freeform-shaped patterns [28] and other geometries. Furthermore, integrating active metasurfaces, such as incorporating liquid crystal [11, 75], multiple quantum well structures [76] and phase change materials [77], could also allow encoders with tunable spectral features to capture more information from light (Fig. 4b) [28, 78]. Based on diverse tuning principles, such as electrical [11], thermal [79, 80], optical [81, 82], mechanical [83], or chemical modulations [84], tunable metasurfaces can adjust their spectral characteristics. This enables spectrometers to be more flexible for measurement tasks like switching detection wavelengths across broad bands or dynamically capture and enhance specific spectral features in spectral monitoring. The dynamic response of tunable metasurfaces offers possibilities for multi-scenario spectral detection. Additionally, their miniaturized and highly integrated





**Fig. 4** Diffraction optical elements of miniaturized reconstructive spectrometers. **a** PC with different geometric characteristics and the corresponding spectral responses. **b** Metasurface combined with external field-induced responsivity. **c** Ultra-simple diffractive structures comprising solely arbitrarily shaped pinhole. **a** Adapted from ref. [69] with permission under CC BY 4.0. **b** Adapted from [11] with permission from Springer Nature. **c** Adapted from ref. [86] with permission under CC BY 4.0

nature makes it promising to incorporate them into portable spectrometers.

With the powerful light field modulation capabilities, metasurfaces enable multidimensional control and detection, offering viable solutions for the parallel measurement of spectra, spatial, polarization, and other dimensions. However, metasurfaces may demonstrate sensitivity to the angle, potentially impacting spectrometer performance [38]. Therefore, precise control over incident light angles during the calibration phase is crucial. Furthermore, in high-speed modulation detection scenarios, it is essential to fully consider the effects of modulation rates and device response times. In addition, tunable devices still encounter challenges such as longevity, stability, and tuning range [85]. When integrating these devices into spectrometer systems, the complexity and cost of manufacturing are also significant. Considering the complexities associated with the fabrication of high-precision nanophotonic structures, studies have also adopted ultra-simple diffractive structures

comprising solely an arbitrarily shaped pinhole [86, 87], see Fig. 4c.

#### 2.2.4 Band-gap modulation

Low-dimensional materials, including zero-dimensional quantum dots (QDs), one-dimensional transverse nanowires, and two-dimensional materials (2DMs), have garnered significant attention in spectrometry as their bandgaps can be engineered by compositional or structural changes.

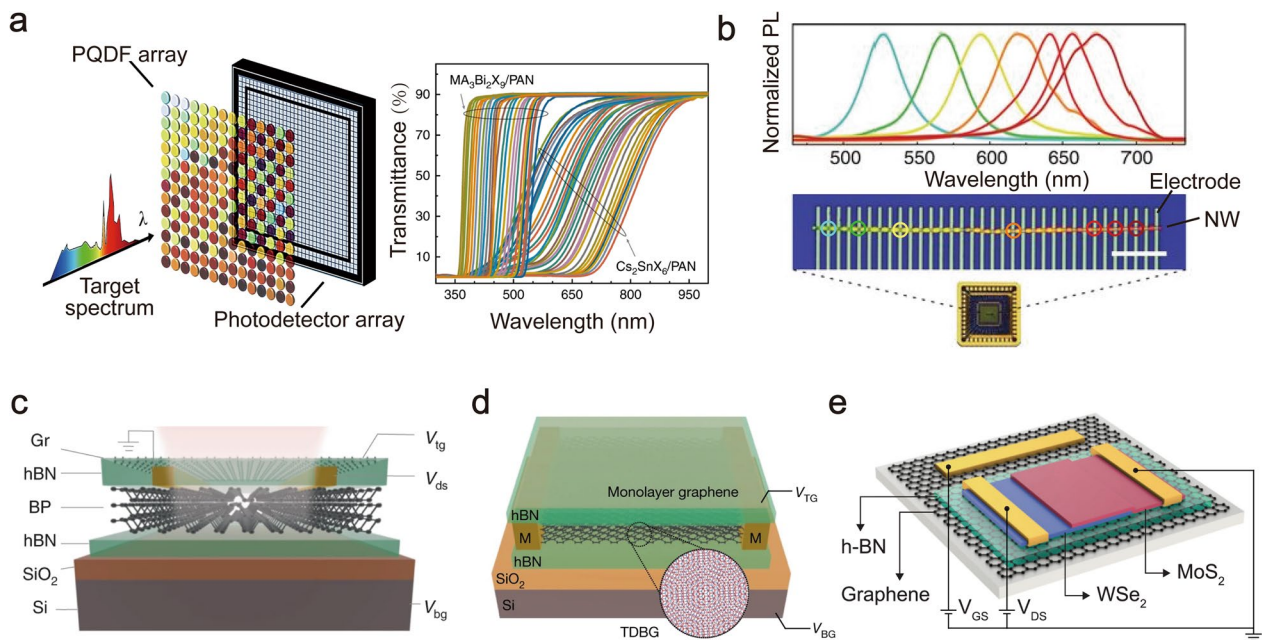
In 2015, Bao and Bawendi [88] demonstrated the potential of colloidal quantum dots (CQDs) for spectrometer applications by precisely engineering their absorption properties through composition, size, and shape control (Fig. 5a). While their work significantly advanced the field and popularized computational spectrometer designs based on low-dimensional materials, this approach still required an external detector array, limiting further miniaturization [89–92].



In 2019, Yang et al. [16] demonstrated a nanowire (NW)-based spectrometer in which the detector itself possessed inherent spectral modulation capabilities (Fig. 5b). The compositionally graded NW incorporated 38 distinct units along its axis, each exhibiting tailored spectral response, thereby eliminating the need for a separate modulation component or external detector array. Further miniaturization requires a single detector capable of generating multiple spectral modulation functions, eliminating the need for detector arrays while maintaining reconstruction accuracy. Following this principle, 2DM-based spectrometers with time domain sampling (i.e., from multiple states with one channel) have emerged as a viable solution. This approach has been realized through three distinct photodetector configurations: mono-2DM [17, 93] (Fig. 5c), twisted bilayer 2DM [94] (Fig. 5d), and van der Waals heterostructures [18–20, 95, 96] (Fig. 5e). These systems electrically tune spectral response by leveraging phenomena such as the Stark effect in black phosphorus (BP), the moiré superlattice effect in twisted bilayer graphene, and the transport-mediated photocurrent assisted by interlayer excitons or the inelastic band-to-band tunneling in heterojunctions. The principle behind 2DM-based spectrometers involves the modulation of energy band structures and photoresponse by applying an external electric field.

Furthermore, leveraging the non-volatile optoelectronic memory effect [97] induced by interface trap states in van der Waals heterostructures (vdWHs), researchers have developed functional devices that may enable the integration of spectral sensing, storage, and processing modules [21]. These innovations continue to push the boundaries of low-dimensional ultra-miniaturized spectrometry toward the realization of spectrometer-on-a-pixel.

In practical applications, tunable reconstructive spectrometers offer certain advantages and disadvantages over spatially arrayed reconstructive spectrometers. For the latter, the angular sensitivity and polarization sensitivity of certain devices increase the precision requirements of the calibration setup, adding to the calibration cost. Additionally, spatial misalignment of components and the diffraction effect introduce spatial crosstalk, which is also an issue for such systems. In contrast, tunable reconstructive spectrometers, with their single-detector architecture, eliminate the influence of spatial crosstalk and enable the development of more compact spectral detection systems. However, the similarity of spectral responses across different tuning states may undermine the reconstruction accuracy. Furthermore, such devices encounter higher operational overhead to maintain system stability under dynamic environmental conditions and face



**Fig. 5** Band-gap tunable optical elements of miniaturized reconstructive spectrometers. **a** CQDs and the transmittance curves varying with components. **b** Real-color photoluminescence (PL) image and corresponding spectra of gradient NW. 2DM composed by **c** mono-2DM, **d** twist bilayer 2DM, and **e** vdWH. **a** Adapted from Ref. [90] with permission under CC BY 4.0. **b** Adapted from ref. [16] with permission from AAAS. **c** Reprinted from Ref. [17] with permission from Springer Nature. **d** Reprinted from Ref. [94] with permission from Springer Nature. **e** Reprinted from Ref. [18] with permission from AAAS

significant challenges in large-scale manufacturing. For example, in spectral devices utilizing BP, which is a material prone to degradation in ambient conditions, the narrow bandgap and long-range carrier interactions impose stringent constraints on operating temperature [17].

### 2.3 Decoding algorithm

Intuitively, Eq. (2) can be solved with direct matrix inversion or via least squares fitting. In practice, the inversion problem formulated above is typically ill-posed [9], rendering the naive solution noise-sensitive and unstable. The underdetermined linear equation system aims to increase the compression ratio, which demands the development of new spectral recovery algorithms [16, 18, 19]. Recent proposals in the literature include iterative and neural network-based methods for precise, high-compression spectral reconstruction [98, 99].

#### 2.3.1 Iterative reconstruction algorithms

Due to the underdetermined nature of Eq. (2), the inversion process is typically transformed into the following optimization problem:

$$\min_S \|\mathbf{R}\mathbf{S} - \mathbf{I}\|_2^2 + \gamma \mathbf{L}(\mathbf{S}) \quad (4)$$

Here,  $\mathbf{S}$ ,  $\mathbf{R}$ , and  $\mathbf{I}$  are consistent with their definitions in Eq. (2), and  $\gamma$  is a regularization parameter that can be determined by the L-curve criterion [100] or the generalized cross-validation method [101]. As a regularization

term,  $\mathbf{L}$  is used to enforce prior knowledge into the sought spectrum, thereby improving the solution of the inverse problem. For example, total variation (TV) [102] or the  $L_2$  norm (Tikhonov norm) [103] can help to smooth the reconstructed results, the nuclear norm can promote low-rank [104] solutions, and the  $L_1$  norm,  $L_p$  norm ( $0 < p < 1$ ) or TV can encourage sparsity in the solutions [105]. Table 1 summarizes several widely used algorithms based on Eq. (4), highlighting their key characteristics and applicable scenarios. Thus, we can carefully choose the regularization term to embed desired characteristics and improve the accuracy of the reconstructed spectrum [106, 107]. For instance, in fluorescence spectroscopy or the detection of quantum dots, which exhibit smooth spectral characteristics, the introduction of  $L_2$  regularization proves to be effective [108]; In chemical molecular detection, where certain gas molecules possess sharp absorption peaks, TV regularization is a suitable choice; In hyperspectral imaging tasks, the non-local similarity of spectral images introduce low-rank properties, making regularizations such as nuclear norm highly recommended. It should also be noted that in scenarios demanding real-time detection, algorithms characterized by greater simplicity and computational speed exhibit distinct advantages, whereas in the presence of significant measurement noise, the robustness of the algorithm emerges as a critical evaluation metric. Although we do not delve into algorithmic variants or advanced modifications, incorporating application-specific spectral priors into reconstruction algorithms is a critical approach to strengthening the connection between algorithm and spectrometer performance.

**Table 1** Commonly used algorithms for spectral reconstruction tasks.  $D$  is a prior constraint matrix such as a difference operator or a smoothing matrix, among others.  $D_1$  denotes the first-order difference operator

| Algorithm           | General form                                                                                                                                                 | Characteristic                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Pseudoinverse       | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2$                                                                                                           | High demands on the properties of matrix $\mathbf{R}$<br>Sensitive to noise |
| LASSO               | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2 + \gamma_1 \ \mathbf{S}\ _1$                                                                               | Effective for sparse signal<br>Strong robustness against noise              |
| Tikhonov            | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2 + \gamma_2 \ \mathbf{S}\ _2^2$                                                                             | Effective for smooth and continuous broadband signals                       |
| Elastic-Net         | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2 + \gamma_1 \ \mathbf{S}\ _1 + \gamma_2 \ \mathbf{S}\ _2^2$                                                 | Effective for sparse and continuously smooth signals                        |
| Quadratic variation | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2 + \gamma_1 \ \mathbf{D}_1 \mathbf{S}\ _2^2$                                                                | Conducive to Gaussian-like signal reconstruction                            |
| General Tikhonov    | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2 + \gamma_1 \ \mathbf{D}\mathbf{S}\ _2^2$                                                                   | Signal-dependent $D$ selection<br>Enhanced universality                     |
| Total variation     | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2 + \gamma_1 \ \mathbf{D}_1 \mathbf{S}\ _1$                                                                  | Effective for piecewise smooth signals<br>Preserving edge information       |
| Fused LASSO         | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2 + \gamma_1 \ \mathbf{S}\ _1 + \gamma_2 \ \mathbf{D}_1 \mathbf{S}\ _1$                                      | Effective for smooth signal with certain sparsity                           |
| Elastic-D1          | $\min_S \ \mathbf{R}\mathbf{S} - \mathbf{I}\ _2^2 + \gamma_1 \ \mathbf{S}\ _1$<br>$+ \gamma_2 \ \mathbf{S}\ _2^2 + \gamma_3 \ \mathbf{D}_1 \mathbf{S}\ _2^2$ | Suitable for complex mixed signals with diverse characteristics             |

Besides, we can obtain the sparse representation of  $\mathbf{S}$  by selecting an appropriate sparse basis and combining Eqs. (3) and (4). Then, the optimization problem defined in Eq. (4) can be converted to:

$$\min_{\alpha} \|\mathbf{R}\varphi\alpha - \mathbf{I}\|_2^2 + \gamma \mathbf{L}(\alpha). \quad (5)$$

Common sparse bases, such as Gaussian bases, have proven to be effective for spectral reconstruction. However, the limited flexibility of these pre-defined sparse bases may hinder the reconstruction performance, particularly for more complex and uncommon spectral data. Therefore, some data-driven methods, like dictionary learning [109], have been proposed, which support the customization of sparse bases from the existing training data. Once the sparse bases and regularization terms are determined, the optimization problem Eq. (5) can be solved using iterative algorithms such as greedy [110–113] and convex optimization [114–116] algorithms.

Note that nearly all the algorithms mentioned above belong to iterative reconstruction algorithms, and their intricate iterative nature requires enormous computational resources, preventing their use in scenarios requiring rapid or even real-time spectral sensing. Although dictionary learning has shown improvement in reconstruction speed, it is essentially a linear model, which may be insufficient to capture the complex nonlinear characteristics and variations in spectral data. A nonlinear and computationally efficient model is more appealing, and the neural network introduced in the following subsection is well-suited for this purpose.

### 2.3.2 Rapid inversion via deep learning algorithms

A neural network is an assembly of layered mathematical operations that mimics the structure and functionality of biological neural networks. It consists of a large number of interconnected artificial neurons, which are mathematical functions that take one or more input values, perform a non-linear operation on the weighted sum of these inputs, and produce an output value. During the training stage, the input data is encoded by the neural network during forward propagation, and the network continuously updates its parameters through the backpropagation process by minimizing a loss function, finally allowing it to perform tasks accurately according to requirements. The loss function typically represents the deviation between the output of the network and the expected output (ground truth). Additional regularization terms can be incorporated to impose constraints on the output data which is similar to the case in Eq. (4). Despite the requirement for large-scale training data and extensive computational resources during the training phase, the neural network enables rapid and accurate

reconstruction of spectral information once the training is finalized.

In the task of spectral reconstruction, Multilayer Perceptron and Convolutional Neural Networks (CNNs) have been widely applied. While purely data-driven methods are prone to overfitting, incorporating domain-specific constraints can mitigate unphysical solutions and improve computational efficiency. Therefore, incorporating a physical model for preprocessing the measurements [89, 117, 118] or introducing regularization terms in the loss function [119] is often a more effective strategy. In addition, due to the scarcity of large-scale spectral datasets, small-data learning theory [120] and the unfolded networks, also known as the plug-and-play framework [121–123], may also provide important directions for future development of spectral reconstruction techniques [124, 125]. In spectral reconstruction tasks involving image processing, network architectures such as U-Net [119, 126] and Transformers [98, 127] have achieved outstanding results due to their excellent feature extraction capabilities. Moreover, a new concept called ‘neural spectral fields’ (NSF) has recently been proposed to reconstruct the spectrum [23]. The proposed method is implemented on a spectrometer platform with bandgap modulation. In their work, the authors trained a feature extractor to capture deep features from the measured current–voltage curves. These curves exhibit unique variations based on the characteristics of the spectrum, such as specific turning points or distinctive shapes, enabling the extraction of features that represent the spectral properties. The extracted features, combined with wavelength information encoded through positional encoding, are then fed into a multilayer perceptron, which decodes the input into a continuous spectral function. Finally, the reconstructed spectrum is refined by imposing physical constraints to ensure accuracy. Compared with traditional spectral reconstruction algorithms, which produce discrete spectrum, the NSF could generate continuous spectral functions, giving it an advantage in high-resolution spectral reconstruction. Moreover, the feature extractor and physical constraints enhance the model’s generalization beyond purely data-driven algorithms and boost the accuracy of spectral reconstruction. Many of the aforementioned models, despite their good performance in reconstruction, are relatively complex and difficult to support on portable devices. To achieve computational resource deployment on such platforms, a lightweight model based on binarization, BiSRNet [40], may be a viable solution.

### 3 Design of encoders

As a critical determinant of spectral reconstruction quality, encoder design remains a central optimization challenge in miniaturized spectrometers. The ideal encoder must simultaneously satisfy two requirements: (1) establishing orthogonal spectral response patterns to maximize information encoding capacity, and (2) translating the designed response characteristics into physically realizable devices. These requirements have directed research efforts into two complementary approaches: forward modeling (predicting spectral responses from known device parameters, e.g., structure and composition) and inverse design (deriving optimal structures for target spectral responses). Although these two approaches have not yet been widely adopted in spectrometer design, they have been extensively applied in the design and response prediction of various photonic structures. Given their exceptional capability to establish structure-response relationships, we posit that these techniques can be readily adapted to spectrometer design. While comprehensive reviews have thoroughly documented these methodologies and their recent implementations [128–137], our discussion is confined to their fundamental conceptual principles, aiming to provide insights for spectrometer design.

Moreover, the performance of a reconstructive spectrometer depends not only on the spectral response matrix (determined by the hardware encoder) but also on the inversion algorithm. This dual dependency significantly increases the complexity and workload of designing a spectrometer system compared to focusing on just one of these components. The emerging end-to-end design paradigm [34] offers particular promise by co-optimizing physical encoder configurations with algorithmic reconstruction processes through unified computational frameworks. This holistic approach addresses the fundamental challenge of aligning hardware capabilities with computational requirements, a critical alignment often overlooked in component-focused designs. Notably, despite its transformative potential for spectrometer development, current literature lacks substantive discussions on end-to-end system optimization strategies, a gap this review specifically aims to address through subsequent analysis.

#### 3.1 Forward modeling

Forward modeling typically involves predicting the optical characteristics of a given photonic structure, with a special focus on spectral response properties. Due to the intricate light-matter interactions, this mapping process often necessitates complex numerical simulations. In conventional methods such as finite-element method [138, 139], finite difference time domain method (FDTD)

[140], rigorous coupled wave analysis (RCWA) [141, 142], and transfer matrix method (TMM) [143, 144], the calculation process is typically time-consuming. Fortunately, the development of neural networks has significantly enhanced the efficiency of forward modeling. By incorporating the physical parameters of the device or the images of each unit cell in periodic structures as inputs, neural networks can serve as surrogate models to predict the device's optical responses (such as absorption/transmission spectra) and compare them against predefined labels. Subsequently, through backpropagation, the network parameters are updated, thereby enabling the trained neural network to predict the optical response of a given device.

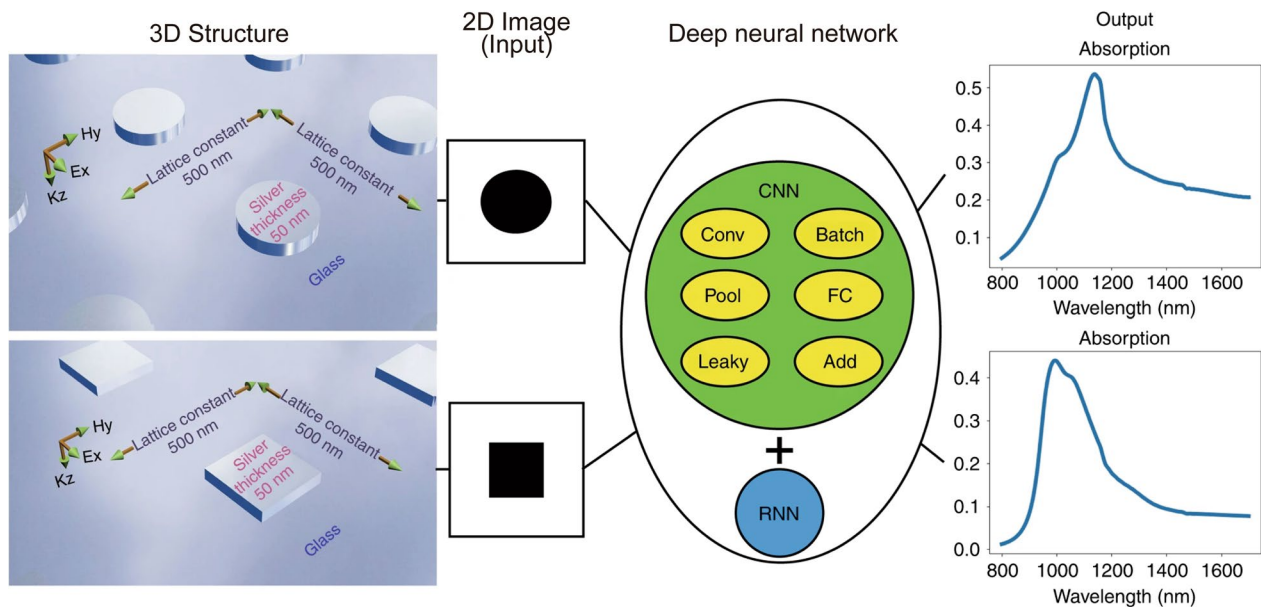
Figure 6 presents a study that predicts the absorption spectrum of plasmonic structures from their 2D images [145]. By flattening the 3D structure and combining CNNs with recurrent neural networks (RNNs), the trained surrogate model can output predictions with high accuracy. In addition, the surrogate model is also capable of predicting transmission and reflection spectra [146], as well as inferring spectral information of similar structures [147]. It can even uncover the impact weights of structural parameters on the response [148] and the latent design space [149]. Surrogate models may not completely surpass traditional Maxwell solvers in terms of accuracy, but they typically exhibit excellent predictive accuracy and computational speed [145, 150, 151]. Therefore, in specific design tasks, forward modeling based on neural networks offers a superior choice.

However, generating large-scale training data through numerical simulation is time-consuming, and the accuracy and robustness of the model largely depend on the training data's quality. Physics-Informed Neural Networks (PINNs), as a type of neural network for solving nonlinear partial differential equations, may offer a potential solution [39, 152, 153]. By incorporating physical laws into the loss function, PINNs can learn the physical constraints between structure and response, thereby enabling the development of models with greater generalization capabilities using fewer training data. Considering the inherent demand for training data in neural networks and the challenges associated with model convergence during the training process, publicly available datasets or surrogate models are highly beneficial for forward modeling [131, 135, 137].

#### 3.2 Inverse design

Inverse design is an increasingly popular approach in the field of nanophotonics and photonic device design. Contrary to forward modeling, the inverse design aims to generate the desired device structures with tailored optical responses. This approach often functions as an





**Fig. 6** Forward modeling of various neural network architectures. Image processing can predict the absorption curve from the 2D image of a given structure. The figure illustrates two distinct structures serving as inputs to neural networks, resulting in different absorption curves, where the neural networks are a combination of CNN and RNN. Reprinted from Ref. [145] with permission under CC BY 4.0

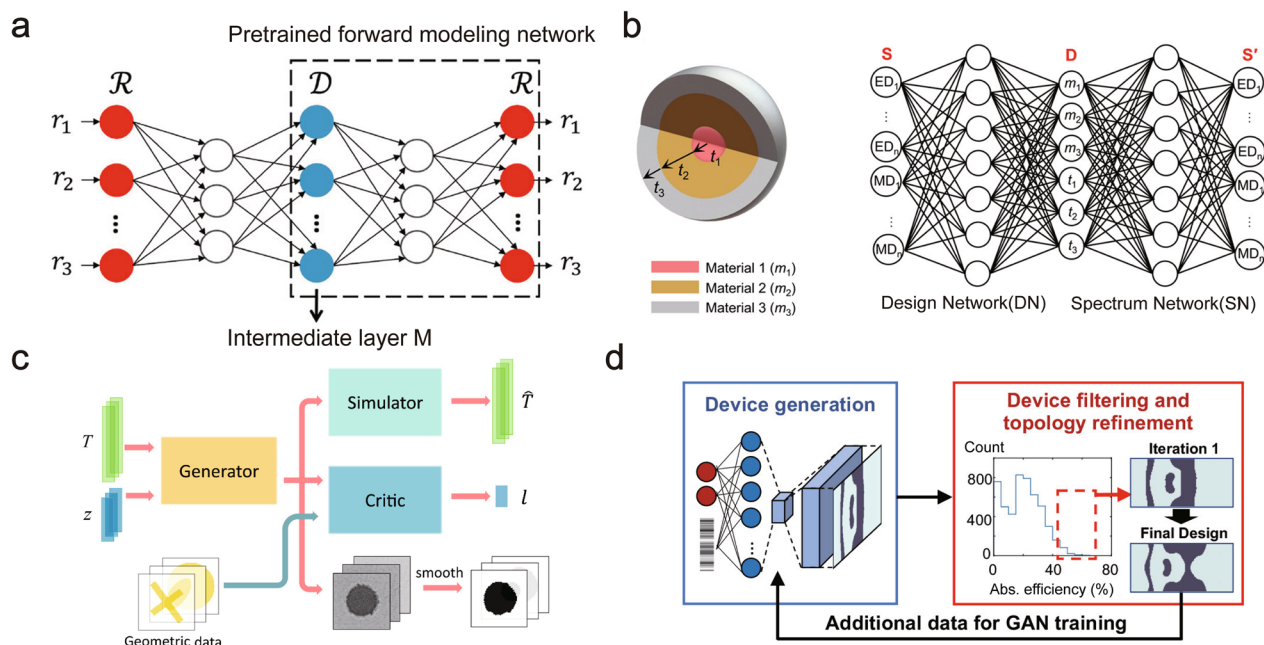
inverse modeling tool, establishing a direct mapping between target optical behaviors and the corresponding device architectures. In addition, when metrics such as efficiency are prioritized, inverse design enables an objective-driven framework for structural refinement. This framework iteratively adjusts design parameters to ensure the figure of merit (FOM) meets the predefined performance targets [44, 154–156].

### 3.2.1 Inverse modeling

In the context of forward modeling, the response corresponding to a nanostructure can be determined by solving Maxwell's equations using numerical methods. Conversely, in inverse modeling, the mapping from response to nanostructures is often a complex, nonlinear transformation without explicit guiding equations. Neural networks, however, can learn this implicit nonlinear mapping in a data-driven manner, making them a powerful tool for inverse modeling. A common challenge in inverse modeling is the one-to-many problem, where a single optical response may correspond to multiple structures. This ambiguity can lead to difficulties in converging the inverse model. Initially, this issue was addressed by combining forward modeling and inverse design in a tandem architecture [29], where the structural parameters generated by the inverse network were fed into a pre-trained forward network to obtain predicted optical responses (see Fig. 7a, b). The loss function was defined as the mean square error between input and output

responses, effectively addressing the non-convergence issue by circumventing the non-uniqueness of photonic structures. As research has progressed, various generative models, including generative adversarial networks (GANs) [157–159], have been implemented in inverse modeling, demonstrating promising performance in the prediction and generation of structures.

Generative Adversarial Networks are generative models comprising a generator and a discriminator. The generator aims to produce new samples that resemble real data, while the discriminator tries to distinguish between samples created by the generator and real data. These two models are trained in an adversarial manner, allowing the generator to continuously improve the quality of the produced samples, while the discriminator enhances its ability to discriminate between real and generated samples. Ultimately, the generator is capable of producing highly realistic samples that are difficult for the discriminator to distinguish [159, 161, 162]. In the context of designing metasurfaces [157], a GANs-based system includes a Generator, Critic, and Simulator, as illustrated in Fig. 7c. The Generator takes the target spectrum and noise as inputs to produce the desired geometric patterns. The Critic and Simulator then evaluate these patterns to measure both the distribution distance between the generated and real patterns and the discrepancy between the spectral response of the generated structure and the input spectrum. The network is subsequently optimized by minimizing the outputs of the critic and



**Fig. 7** Schemes of inverse modeling. **a** tandem network for solving the one-to-many problem. **b** Deep-learning-assisted inverse design of core-shell nanoparticles. **c** Generative model for inverse design of metasurfaces. **d** The data generated by the generator can be fed back to the neural networks for retraining. **a** Reprinted from Ref. [29] with permission from ACS. **b** Reprinted from Ref. [160] with permission under CC BY 4.0. **c** Reprinted from [157] with permission from ACS. **d** Reprinted from [159] with permission from ACS

simulator, ultimately enabling the direct generation of structures from the specified spectrum. Furthermore, the patterns generated by the generator can serve as additional training data after topological refinement, reducing the dependence of the neural network on training datasets, as shown in Fig. 7d. In addition to GANs, variational autoencoders, as another type of generative model, have also achieved promising results in inverse modeling [163].

### 3.2.2 Inverse design for device optimization

When optimizing device structure, the primary goal is to ensure that the FOM of the designed device aligns with predetermined performance specifications. This design approach, paradoxically, operates in a forward direction: it initiates by calculating the FOM of an existing structure and then iteratively adjusts the structural parameters according to the FOM, until the final structure meets the desired criteria.

To accomplish this, two primary strategies exist. Firstly, evolutionary algorithms such as genetic algorithms [164] or particle swarm optimization (PSO) [165] can be employed, progressively refining the candidate designs through evolutionary iterations. Genetic algorithm is a typical heuristic method that simulates the biological evolution process to randomly test and iteratively optimize the initial distribution of solution space,

gradually refining the solutions to the problem. Particle swarm optimization is an optimization algorithm based on swarm intelligence, possessing strong global search capabilities and fast convergence rates. Due to the high computational cost of solving Maxwell's equations, these methods require testing a large number of solutions to find satisfactory optimization results, thus incurring significant computational overhead. Alternatively, gradient-based optimization methods, such as adjoint method [130] or level set methods [166], could be used to utilize the gradients of the FOM with respect to device structural parameters to guide parameter optimization. The level set algorithm is a numerical method based on partial differential equations, used to describe the contour changes of objects with respect to specified variables. The adjoint method, by solving the adjoint equation, can compute the derivatives of all design parameters regardless of their quantity, thus transforming complex multivariable optimization problems into a more computationally efficient form.

For the inverse design of photonic devices, the following steps are typically used: firstly, it is essential to define the FOM based on requirements, such as maximizing transmittance or minimizing losses. Subsequently, the device model is constructed. On one hand, the device structure needs to be modeled and initialized, which can be described with a few parameters to fully determine

the device configuration, or the geometric shape can be converted into pixel-like elements to enhance design flexibility. On the other hand, it is essential to model the interaction between the light field and the device, calculate the FOM based on the physical model, and iteratively optimize the FOM using algorithms, continuously updating design parameters until meeting the requirements. Besides, considering the feasibility of the device, employing quantization algorithms to regularize the device structure or using convolution filters to blur small structural features are effective strategies [136], despite inevitably influencing the FOM. In the design of reconstructive spectrometers, the performance often necessitates the evaluation through computing the responses of a set of devices (or different states of one device), leading to higher computational complexity. Furthermore, the lack of a comprehensive evaluation system for such spectrometers poses a challenge for defining the FOM.

In the work of on-chip spectrometers [44], the inverse design of stratified waveguide filters was accomplished using PSO. By utilizing the autocorrelation of individual filter responses and cross-correlation between different filters as the FOM for optimization, their design methodology enables the generation of a minimally correlated filter set. This approach effectively suppresses spectral crosstalk between adjacent channels, thereby enhancing the overall spectral resolution of the system. Similar approaches have also been applied in reconstructive spectrometers enabled by a film encoder [167]. For the inverse design of spectrometers relying on scattering from disorders [66], Maxwell's equations are employed as physical constraints for structure generation. After determining the characteristic modes of the output port, structures that meet the constraints are iteratively screened, ultimately yielding a solution that satisfies the requirements. Despite the relative scarcity of inverse design of spectrometers based on gradient-based methods, related methodologies have been applied to design other photonic devices, including metasurfaces [168–170], metalenses [171], waveguides [172] and photonic crystal structures [173].

The evolutionary algorithms are capable of comprehensively exploring the solution space, thereby holding the promise of finding the global optimum. However, this comes at the cost of a longer optimization process. For gradient-based optimization algorithms, the adjoint method facilitates the computation of gradients, enhancing computational efficiency. Yet, the inherent nature of gradient algorithms makes them prone to converging to local optima. Both methodologies iterate in a loop through forward modeling, ultimately accomplishing inverse design. Thus the speed of forward modeling largely dictates the computational speed of the

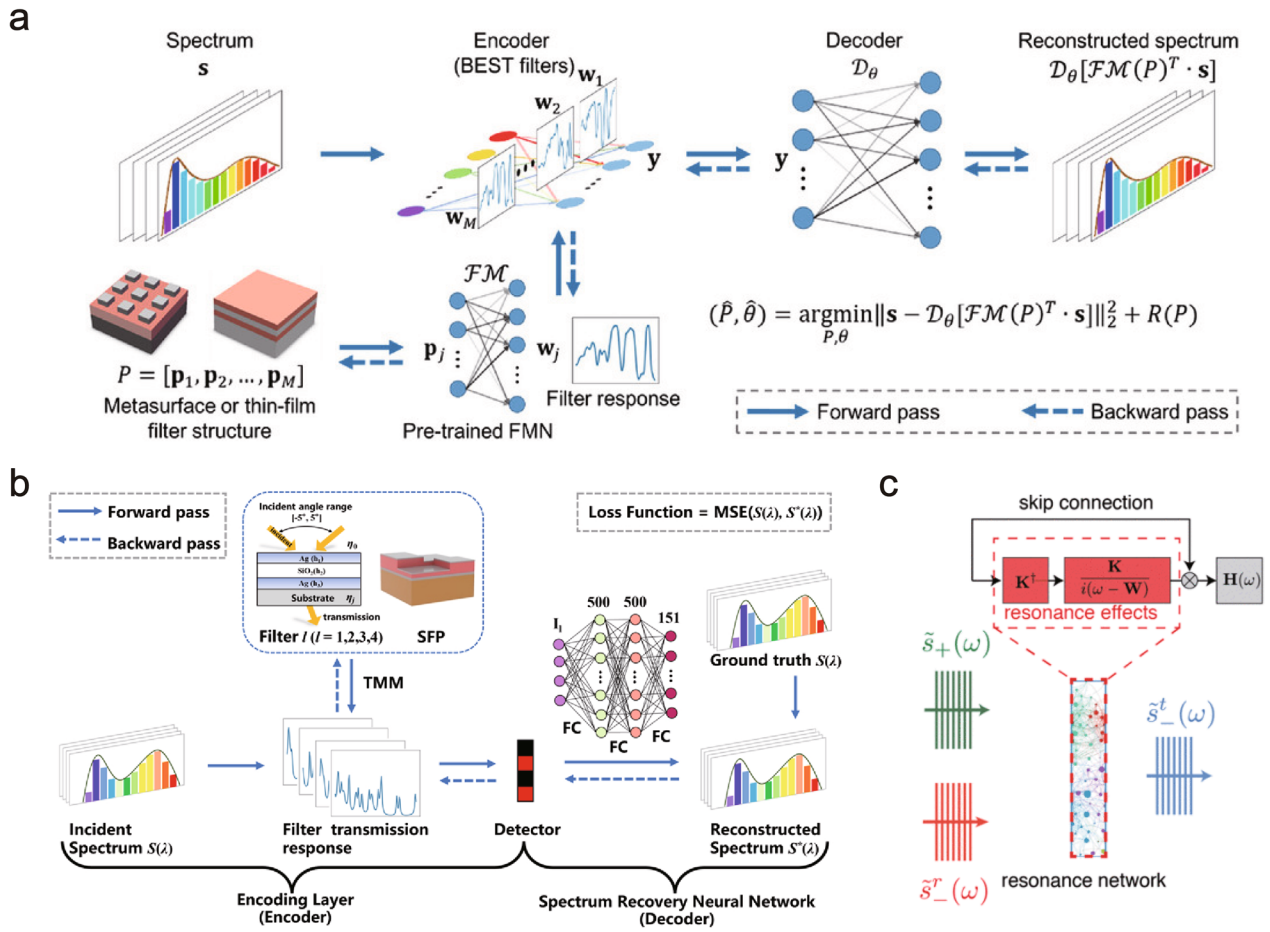
algorithms. Currently, there are relatively few reports on the inverse design of spectrometers, likely due to computational complexity. Therefore, training a neural network as a surrogate model for the forward process to reduce the computational resource required in each iteration would be beneficial for inverse design.

### 3.3 End-to-end design for spectrum system

In previous discussions, the encoder and backend reconstruction algorithms (i.e., the decoder) were regarded as separate entities. Although CS theory establishes a conceptual bridge between the two, the algorithms themselves have not been directly considered during the device optimization process, leading to obstacles towards achieving encoder-decoder coupling.

The end-to-end design paradigm is highly dependent on neural networks, which can comprehensively model task scenarios and leverage backpropagation to optimize various components of the system. This approach has been widely adopted in the field of computational optics [34–37]. For reconstructive spectrometers, end-to-end design necessitates the construction of a physical model based on the encoding process, as well as the development of a forward model to predict spectral responses from device structure and a decoder to reconstruct the spectrum, as shown in Fig. 8a. Ensuring the entire model is differentiable or at least allowing for gradient computation is crucial for effective backpropagation. During training, known spectra are fed into the neural network as inputs. These spectra are encoded according to the spectral responses generated by the device structure, and then reconstructed through the decoder. By evaluating the discrepancies between the reconstructed spectra and the original inputs, a loss function is formulated. Through backpropagation, this process simultaneously optimizes both the decoder parameters and the device structure, enhancing overall performance. Compared to inverse design, this approach minimizes reliance on heuristic tuning of the FOM and explores optimal design configurations within a larger solution space.

This integrated co-optimization fundamentally differs from inverse design approaches, where hardware structures and software algorithms are developed sequentially rather than concurrently. The difference between the two lies in the fact that in end-to-end design, the optical element structure undergoes updates during backpropagation, which in turn modifies the spectral response based on structural changes. This ensures the iterative updates of the spectral response are always constrained by the physical structure during the training process. Within the sequential design paradigm, a series of spectral response functions with desirable properties can be designed based on the orthogonality of the spectral response.



**Fig. 8** End-to-end design for spectrum system. **a** Schematic of the parameter constrained spectral encoder and decoder design framework. **b** End-to-end design framework which can consider different incident angles. Under this framework, the SFP filter array and the decoder are jointly designed. **c** Schematic of trainable coupled resonance layer in d-ALFRED. **a** Reprinted from Ref. [174] with permission from John Wiley and Sons. **b** Reprinted from Ref. [38] with permission from OSA. **c** Reprinted from Ref. [175] with permission under CC BY 4.0

However, the structures obtained by inverse design based on these spectral responses may not be physically realizable. In other words, it is difficult to judge the feasibility of the generated structures through their spectral response alone. By restricting the range of variation of structural parameters or adding other structural constraints, for example, limitations or tolerances of the fabrication process, end-to-end design can ensure structural feasibility, although this may inevitably lead to some loss to the spectral orthogonality.

In constructing an end-to-end network forward model, there is a significant reliance on the nonlinear mapping capacities of neural networks. As mentioned earlier, this imposes stringent requirements on the quantity and quality of the training data. To address this problem, incorporating physical information into the forward model can alleviate the heavy demand for data volume while simultaneously mitigating overfitting issues. Physical models of relatively simple structures, such as MTF

can be directly embedded into the forward model. An encoded filter composed of a superposition Fabry-Pérot (SFP) resonance cavity with the learning of the parameters of the SFP encoder and neural network decoder in an end-to-end manner was presented in Fig. 8b [38]. By incorporating the TMM in the encoding layer, the spectral transmittance of the filter structure can be computed directly using analytical equations, removing the need for a large dataset of structure-transmittance pairs.

Designing metasurfaces poses unique challenges due to the non-differentiable nature of the mapping between structure and spectral response, increasing the integration complexity of physical information while maintaining differentiability for backpropagation. Recent developments like d-ALFRED [175] have helped by embedding differentiable coupled-mode equations, allowing neural networks to learn the mapping between metasurface structures and spectral response (see Fig. 8c), and facilitating the prediction of responses from



structural parameters. Other methods, including the incorporation of automatic differentiation within Maxwell solvers (such as FDTD, RCWA) or the improvement of adjoint methods to solve for gradients [170, 176–178] can achieve differentiability in the backpropagation, facilitating end-to-end design for spectrum systems.

The core of end-to-end design lies in establishing a differentiable forward model, particularly the mapping from structure to spectral properties. Thus, techniques like PINNs used in forward modeling can be adapted for this purpose. Furthermore, the availability of large-scale, open-access datasets correlating metamaterial structures with spectral responses [129, 137] could dramatically accelerate research by reducing reliance on computational or experimental data generation.

Recent studies have begun to focus on integrating automated differentiation with the finite element method, thereby achieving efficient, differentiable solvers for Maxwell's equations [170–172, 176–179]. It is worth mentioning that the use of the adjoint method to calculate gradients for backpropagation in photonics may not be advantageous in the end-to-end design of spectrometer systems. The adjoint method involves an additional electromagnetic simulation to obtain the gradients of the FOM with respect to various input structural parameters. This approach is typically applicable when there are numerous design parameters as inputs and a single scalar FOM as output [180]. However, once the number of output parameters increases, as in the case of spectral responses across multiple channels in a spectrometer system, the computational cost associated with the adjoint method becomes prohibitively expensive.

Beyond computational bottlenecks, another critical challenge in end-to-end design lies in bridging the gap between simulation and fabrication. Zheng et al. [181] introduce the concept of neural lithography, which integrates pre-trained photolithography simulators into optical design frameworks to address deviations between idealized and practical lithographic processes. By employing data-driven methodologies, fabrication feasibility constraints allow for more precise simulation during the design phase, achieving the development of comprehensive and intelligent end-to-end design systems. However, larger dataset requirements, increased sensitivity to data noise, and insufficient theoretical foundations remain critical obstacles. Despite the numerous challenges outlined above, end-to-end design offers an intuitive and powerful solution for future spectrometer design, offering significant guidance for bridging the gap between simulation and real-world scenarios.

## 4 Conclusions and outlook

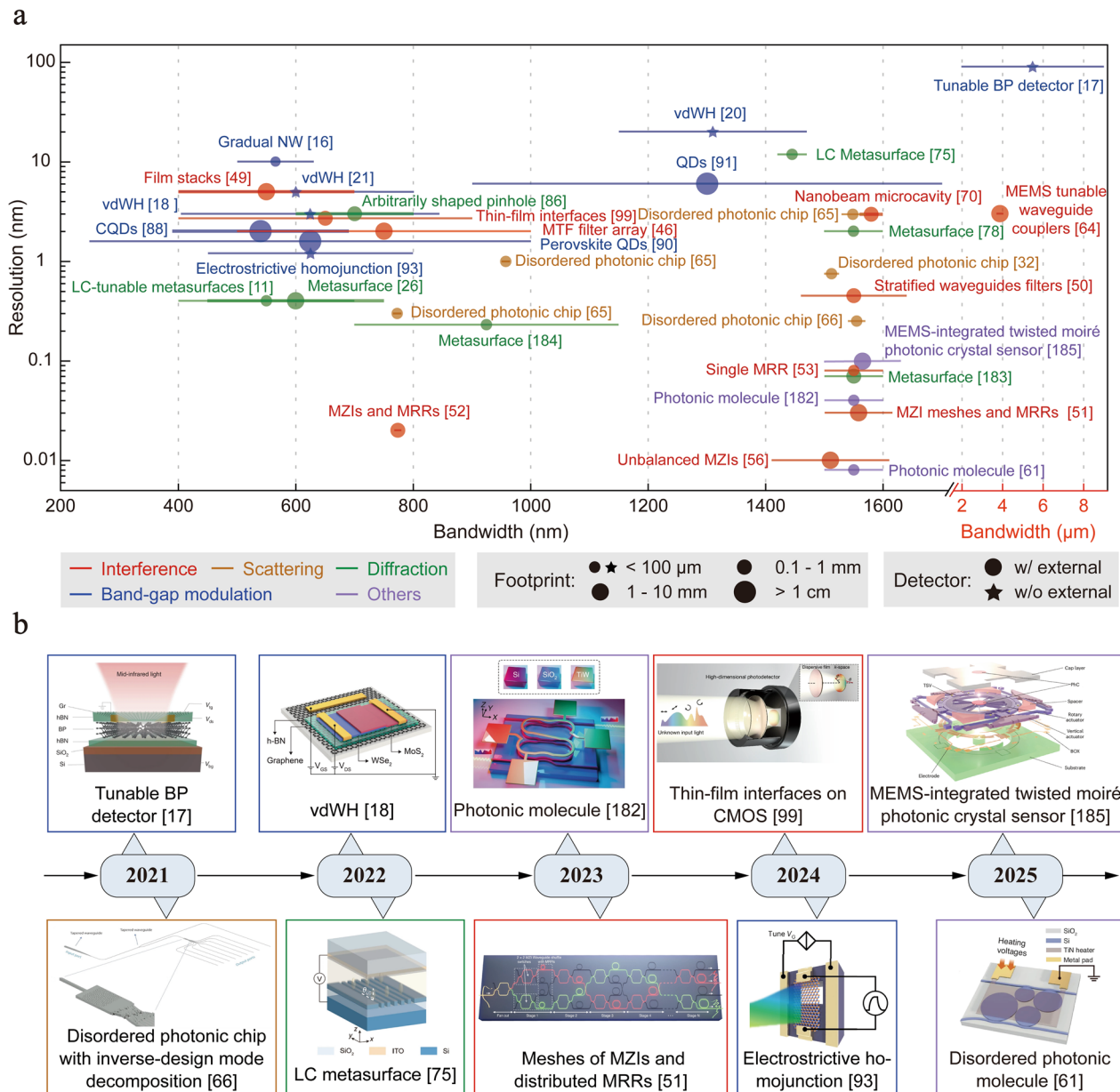
Recent advancements in miniaturized spectroscopic systems have revolutionized traditional spectral analysis paradigms. This review synthesizes interdisciplinary progress in reconstructive miniaturized spectrometers, encompassing mathematical frameworks, hardware innovations, and end-to-end design strategies that unify optical encoding with computational reconstruction. We began by establishing the mathematical framework. Four distinct encoding strategies spanning interference, scattering, diffraction, and band-gap modulation were analyzed, highlighting their unique mechanisms for spectral modulation and trade-offs in resolution, sensitivity, and fabrication complexity. Innovations in decoding algorithms, particularly deep learning-based rapid inversion, were discussed as key enablers of high-accuracy spectral recovery under constrained computational resources. Equally transformative is the emerging paradigm of end-to-end encoder-decoder co-design. By implementing differentiable physical models within neural network frameworks, researchers can achieve joint optimization of spectral modulation elements and reconstruction algorithms with algorithmic robustness. Collectively, these advances are driving the transition from bulky benchtop systems to ultra-compact, field-deployable spectrometers capable of real-time, high-fidelity spectral analysis across diverse applications.

It should also be noted that comprehensive performance evaluation requires consideration of multiple parameters, including bandwidth, sensitivity, dynamic range, and long-term stability. While reconstructive spectrometers excel in miniaturization and cost-effectiveness, their performance characteristics may vary across different operational contexts and are typically optimized for specific use cases. In recent years, an increasing number of novel designs have been proposed to address the inherent trade-offs among physical size, resolution, bandwidth, and spectral channel density in computational spectrometers, yielding successful outcomes [60–62, 182, 183]. The concurrent reduction in footprint and enhancement in resolution with wider bandwidth demonstrate that these devices can achieve some performance metrics (e.g., resolution) comparable to that of conventional systems in targeted scenarios while occupying orders-of-magnitude smaller scales as illustrated in Fig. 9a.

In Fig. 9b we present a timeline of key breakthroughs over the past 5 years, highlighting the rapid advancement in this field. The evolution of spectrometer technology has increasingly emphasized tunability and reconfigurability as essential features for compact, high-resolution, and versatile spectral sensing. This is achieved through strategies that enable dynamic spectral control without relying on fixed optical components. Electrical tuning

of material properties plays a key role, as seen in van der Waals junctions such as  $\text{MoS}_2/\text{WSe}_2$  heterostructures, where gate-tunable interlayer transport enables sub-nanometer resolution from 405 to 845 nm [18]. Similarly, black phosphorus photodetectors use the Stark effect for electrically controlled bandgap tuning in the mid-infrared

(2–9  $\mu\text{m}$ ) [17]. Photonic circuit reprogramming offers another pathway, exemplified by Mach–Zehnder interferometer networks integrated with micro-ring resonators, which use thermo-optic tuning to achieve 30 pm resolution over a 115 nm bandwidth [51].



**Fig. 9** The progress of reconstructive spectrometers in recent years. **a** Comparative analysis of resolution, bandwidth, footprint, and detector characteristics in miniaturized reconstructive spectrometers. The plotted data encompass both devices (color-coded) and reference (numbered in parentheses). Footprint measurements are restricted to optical components essential for light resolution and detection, excluding auxiliary elements such as electronic circuitry or external packaging. Note that the bandwidths for [52, 64, 65] (covered by the data points) are 12 nm, 350 nm, and 15 nm respectively. **b** Timeline illustrating key breakthroughs for reconstructive spectral systems over the past 5 years. Images are reprinted from refs. [17, 51, 61, 93, 99, 182, 185] with permission from Springer Nature, Ref. [66] with permission from John Wiley and Sons, Ref. [18] with permission from AAAS, and Ref. [75] with permission under CC BY 4.0

Another focus of these studies has been to overcome the conventional trade-off between resolution and bandwidth while achieving extreme miniaturization. One approach achieves this by leveraging dispersion-engineered mode splitting in a photonic molecule composed of two coupled microrings [182]. This technique decouples wavelength channels across multiple free-spectral ranges, enabling an ultrahigh resolution of 40 pm over a bandwidth exceeding 100 nm, all within a compact footprint of only  $60 \times 60 \mu\text{m}^2$ . Another notable advancement introduces an innovative design based on an N-body disordered photonic molecule formed by three or more heteronuclear microdisks [61]. By exploiting chaotic electromagnetic coupling, this structure generates quasi-random spectral response matrices. This novel mechanism enables the system to transcend the traditional resolution-bandwidth-size trade-off, achieving an impressive resolution of 8 pm across a bandwidth greater than 100 nm within an ultracompact area of just  $70 \times 50 \mu\text{m}^2$ .

In parallel, significant efforts have been devoted to the computational recovery of multiple dimensions of light [75, 184–189]. These additional dimensions can provide richer and more comprehensive data. For example, polarization information can reveal the orientation and structure of materials [190, 191], which is crucial for applications in materials science, telecommunications, and biomedical imaging. A recent breakthrough demonstration using a uniform dispersive thin film has enabled computational multidimensional sensing, allowing concurrent recovery of polarization and spectral information via integrated physical-algorithmic design. [99]. Continuing the trend towards miniaturization, a compact hyperspectro-polarimetric camera utilizing an encoding metasurface has effectively captured multi-dimensional information in a single snapshot [184]. Similarly, a microelectromechanical system (MEMS)-integrated twisted moiré photonic crystal sensor has demonstrated remarkable miniaturization and adaptability for dynamic multi-dimensional light control and sensing functions [185]. Despite rapid advances, the transition of reconstructive spectrometers from laboratory prototypes to industrial-scale deployment still faces several critical challenges. A key barrier lies in scalable fabrication and cost-performance trade-offs. For high-resolution encoders, current technologies like electron beam lithography encounter significant scalability limitations primarily due to cost constraints. Efforts to reduce costs often involve alternative materials or simplified processes, sometimes at the expense of optical performance, device miniaturization, or long-term reliability. For example, while deep ultraviolet lithography offers a cost-effective alternative, it does so at the expense of poorer resolution. Emerging techniques such as roll-to-roll processing of advanced

polymeric materials offer promising solutions, achieving material cost reductions of over 90% [192].

At the design level, significant technical challenges remain. Minimizing the number of encoding channels is crucial to reduce the dimensionality of the inverse problem and ease the computational burden of spectral reconstruction. This requires optimized optical encoding and the adoption of efficient compressed sensing strategies that capture maximal spectral information with minimal measurements. Developing reliable neural network-based reconstruction models is further constrained by the need for large, high-quality training datasets, which are often difficult and expensive to acquire. Even with robust models, miniaturized spectrometer systems frequently face restricted data bandwidth, creating bottlenecks in transferring and processing high-resolution spectral signals. Addressing these limitations calls for data-efficient acquisition combined with advanced on-device computation. Neuromorphic vision sensors [193], inspired by biological vision, use event-driven sampling and in-sensor computation to minimize redundant data and power consumption, enabling real-time spectral analysis in compact, energy-constrained devices. Additional strategies, such as near-sensor and in-sensor computing with memristive implementations [194–197], further reduce latency and energy consumption by minimizing data transfer between memory and processing units. Direct photocurrent-based material fingerprinting without full spectral reconstruction [108] offers another pathway to significantly cut computational complexity and power demand.

Commercial success depends on the emergence of compelling killer applications that clearly demonstrate value. While reconstructive spectrometers show promise in healthcare, environmental monitoring, and industrial inspection, the technology has yet to secure a breakthrough use case to drive mass adoption. The key to unlocking its potential lies in the consumer market, where economies of scale can significantly reduce costs. If a successful application is identified, it could pave the way for broader adoption across other sectors, making them much more accessible. Encouraging signs are emerging, as high-resolution waveguide-based spectrometers enable precise molecular analysis for medical diagnostics, clinical-grade biomarker detection systems such as glucose monitors advance through validation, and compact energy-efficient designs expand into consumer health markets [198], laying the groundwork for broader adoption in industrial and clinical settings. Notably, leading consumer electronics and smartphone manufacturers are already exploring early implementations of spectral sensing in products including in smartphones. Although these are not focused on spectral measurements, such

devices are being used for ambient light detection to enhance photography, signalling early interest in integrating spectral capabilities into everyday technologies.

To conclude, ongoing advancements in computational decoding schemes, design optimization of hardware decoders, and multidimensional light recovery are paving the way for the next generation of miniaturized reconstructive spectrometers. Most notably, spectrometer-on-a-pixel architectures integrated into arrays or waveguides on monolithic chips hold particular promise. As research continues to evolve, miniaturized spectrometers are expected to become increasingly sophisticated, efficient, and application-focused, accelerating innovation across a broad range of scientific and technological domains.

#### Abbreviations

|       |                                      |
|-------|--------------------------------------|
| 2DMs  | Two-dimensional materials            |
| BP    | Black phosphorus                     |
| BTBT  | Band-to-band tunneling               |
| CNNs  | Convolutional Neural Networks        |
| CQDs  | Colloidal quantum dots               |
| CS    | Compressive sensing                  |
| FDTD  | Finite difference time domain method |
| FOM   | Figure of merit                      |
| FP    | Fabry-Pérot                          |
| GANs  | Generative adversarial networks      |
| MRRs  | Microring resonators                 |
| MTF   | Multilayer thin-film                 |
| MZIs  | Mach-Zehnder interferometers         |
| NW    | Nanowire                             |
| OPD   | Optical path difference              |
| PC    | Photonic crystals                    |
| PINNs | Physics-Informed Neural Networks     |
| PSO   | Particle swarm optimization          |
| QDs   | Quantum dots                         |
| RCWA  | Rigorous coupled wave analysis       |
| RNNs  | Recurrent neural networks            |
| SFP   | Superposition Fabry-Pérot            |
| SiN   | Silicon nitride                      |
| SOI   | Silicon-on-insulator                 |
| TMM   | Transfer matrix method               |
| TV    | Total variation                      |
| vdW   | Van der Waals                        |
| vdWHs | Van der Waals heterostructures       |

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#### Author contributions

All the authors contributed to the preparation of the manuscript. Y.Z. and E.Y. contributed equally to this work. Z.S., T.H. and W.C. conceived the initial ideas. Y.Z. and E.Y. led the figure making and writing process. Z.S., T.H. and W.C. supervised this work. All authors read and approved the final manuscript.

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#### Data availability

Not applicable.

#### Declarations

#### Competing interests

The authors declare no competing interests.

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